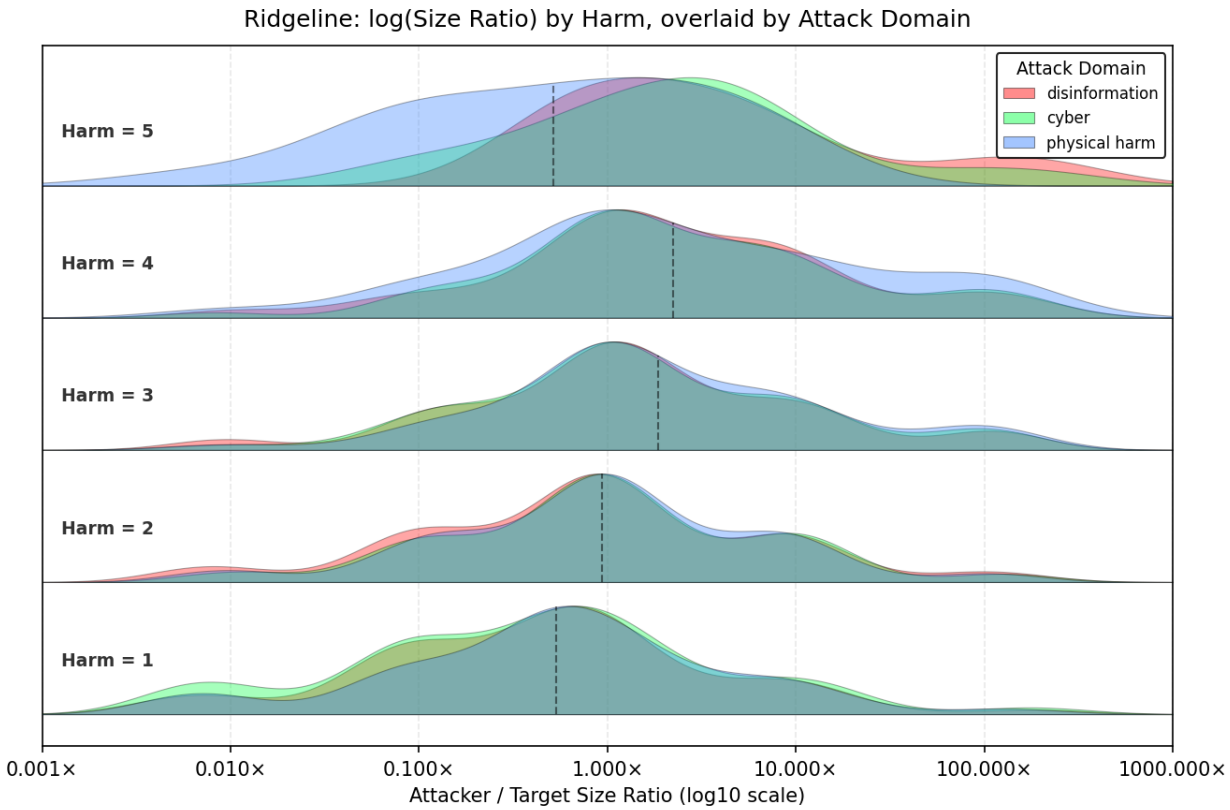

Figure 2: Ridgeline distributions of attacker-to-target size ratios (log scale) stratified by rounded discrete harm levels. Higher harm strata shift toward larger relative attacker sizes, showing that severe jailbreaks are more likely when attackers substantially exceed target scale. Harm dispersion appears similar between attack domain, with stronger differences at harm = 5, especially for physical harm.

Harm variability differed markedly between attackers and targets. Mean harm variance was consistently higher across attackers (0.180) than across targets (0.097), indicating that most dispersion arises from attacker-side behavioral diversity rather than from systematic differences in target defensiveness. This pattern persisted when refusal cases were included or excluded, suggesting that heterogeneity in attacker refusal behavior is the dominant source of harm variability. Refusal frequency was strongly and negatively correlated with mean harm ($\rho = -0.927$, $p < 0.001$), confirming that alignment-driven refusal acts as a key protective mechanism against adversarial success. Detailed variance comparisons, robustness analyses, and per-pair harm distributions are provided in Appendices A–C.

We next discuss the most salient patterns observed across these results and their implications for LLM robustness.

5 Discussion and Implications

Relative model scale and harm outcomes. Figures 1 and 2 show a clear positive relationship between the attacker–target size ratio and mean harm. Larger attacker models tend to elicit higher harm scores from smaller targets, suggesting that relative model capacity systematically influences adversarial success. This pattern implies that expressive and persuasive advantages scale with

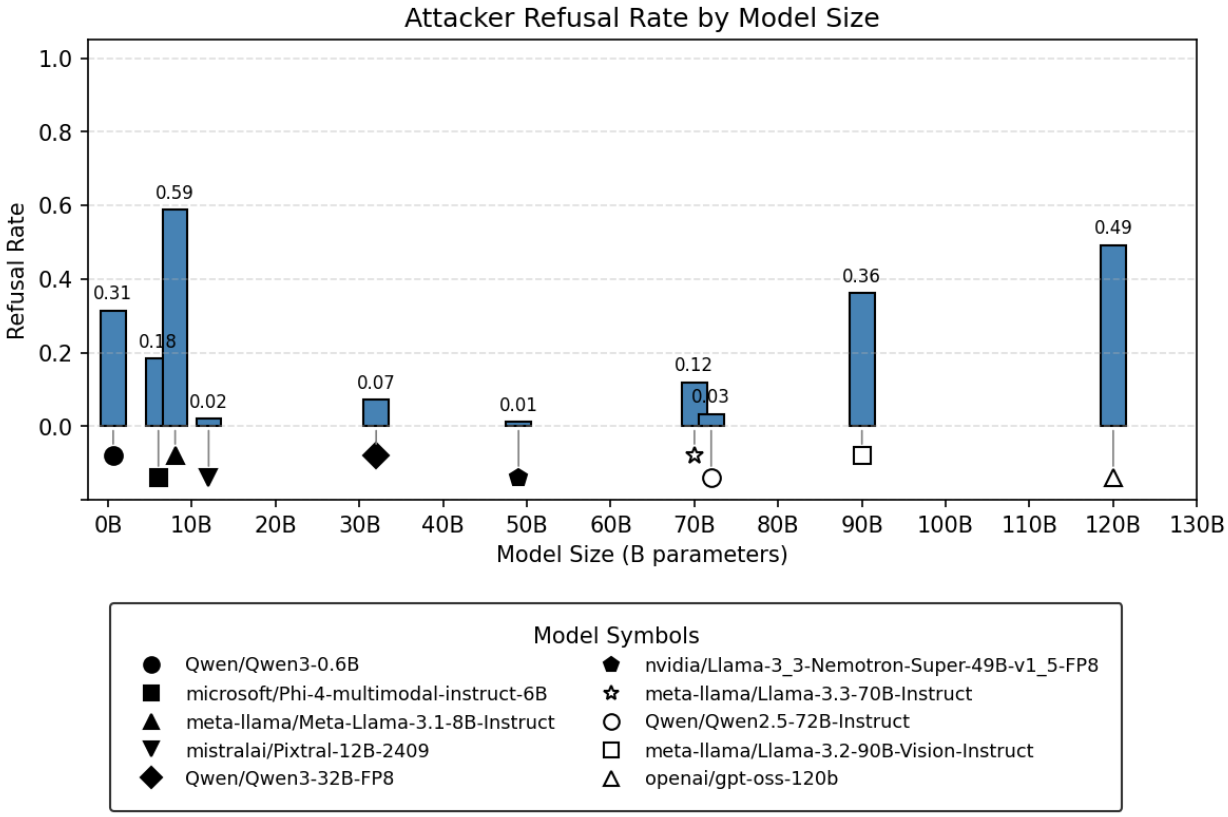

Figure 3: Attacker refusal rates by model size. Refusal frequency decreases with model scale, reflecting stronger adversarial persistence in larger models. The strong negative correlation between refusal rate and mean harm ($\rho = -0.927$, $p < 0.001$) indicates that refusal behavior is a key protective mechanism against harmful outputs.

model size, allowing larger attackers to more effectively bypass target defenses. When attacker and target scales are comparable, harm levels stabilize, indicating more symmetric alignment behavior. These findings point to the importance of *relative* capability—rather than absolute size alone—in determining vulnerability in multi-model interactions.

Refusal dynamics and safety behavior. Refusal dynamics (Figure 3) provide additional insight into behavioral mechanisms. Model size alone is not a strong predictor of refusal rates: some smaller models exhibit high refusal frequencies (e.g., Llama-3.1-8B-Instruct at 59%), whereas several larger models refuse far less often (e.g., Qwen2.5-72B-Instruct at 0.3%). This variability indicates that alignment behavior depends more on data, training, and architecture than solely on size. Nonetheless, the strong negative correlation between attacker-side refusal frequency and mean harm ($\rho = -0.927$, $p < 0.001$) supports the view that refusal remains a critical safety behavior: when attackers abstain, harm remains low; when they persist, harm increases correspondingly. Understanding how alignment reinforcement influences these refusal dynamics is therefore central to improving adversarial robustness.

Attacker-side dominance in harm variance. The mixed-effects analysis (Figure 6) shows that attacker family and size ratio explain the largest share of harm variance, followed by attacker system prompt variant and target family, with minimal contribution from harm domain. This finding suggests that attacker-side characteristics—architecture, scale, and alignment regime—are

the dominant drivers of adversarial outcomes, while prompting strategies and domain factors play secondary roles. Attacker families with stronger persuasive representations appear more capable of eliciting unsafe completions across targets, even when prompt content and task type are held constant. These patterns highlight the need for alignment evaluation that emphasizes attacker-side dynamics rather than target-only robustness testing.

Emergent scaling patterns in adversarial interactions. Taken together, these results provide evidence for *empirical scaling patterns* in multi-model adversarial interactions. Relative model size and refusal behavior jointly shape harm outcomes across architectures, revealing consistent structural regularities in how capability and alignment interact. While not a strict scaling law, this empirically observed relationship illustrates how adversarial vulnerability emerges from asymmetric capability distributions within interacting LLM systems. Future work should extend these analyses to persistent, multi-agent settings to test whether such scaling patterns generalize beyond short-turn interactions and static model pairings.

5.1 Limitations

Correlational rather than causal inference. A central limitation of this study is that its findings are correlational rather than causal. While mean harm increases with the logarithm of the attacker–to–target size ratio, model scale is confounded with multiple other factors, including data quantity, instruction-tuning objectives, and safety alignment procedures. Parameter count therefore serves as a proxy for a broader capability and alignment bundle rather than a pure measure of capacity. As a result, the observed scaling patterns should be interpreted as evidence of association, not proof of direct causal influence. Disentangling these interdependent dimensions will require controlled ablation experiments or counterfactual fine-tuning studies that systematically vary model size while holding training and alignment constant.

Cross-family alignment tuning differences. A second limitation lies in the heterogeneous alignment procedures and safety objectives across model families. Attacker and target systems differ not only in architecture but in the degree and nature of alignment reinforcement—for example, reinforcement learning from human feedback versus rule-based constitutional tuning. These alignment regimes may themselves determine refusal frequency and harm sensitivity, potentially explaining part of the variance attributed to model scale. Because cross-family alignment strength and tuning data remain opaque in most public models, it is difficult to isolate whether higher harm reflects genuine capability differentials or divergent alignment baselines. Future work should incorporate standardized fine-tuning benchmarks or jointly aligned model suites to clarify this distinction.

Simplified interaction structure. The attacker–target exchanges in this study were intentionally simplified to a bounded, few-turn dialogue structure without persistent memory or adaptive planning. While this design allows systematic comparison across many pairings, it abstracts away the richer feedback loops characteristic of real multi-agent systems. In practical deployments, longer or recursive interactions could enable escalation, reinforcement, or emergent coordination between models, amplifying or dampening adversarial effects. Consequently, the observed correlations may underestimate the complexity and persistence of adversarial dynamics in unconstrained multi-model environments.

Model-based evaluation bias. Harm and refusal were judged by large language models rather than human evaluators, introducing potential calibration bias. Automated judging enables scale and consistency, but it also risks circularity: the same architectures used to generate outputs are

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